

HARI SHANKAR BAKKAMANTHULA

Associate AI Automation Engineer

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SUMMARY

Automation Engineer passionate about simplifying complex business processes through AI, automation, and software engineering. I enjoy understanding how organizations work, identifying inefficiencies, and building practical solutions that improve productivity and user experience. A fast learner with hands-on experience across enterprise systems, CRM platforms, full-stack development, and intelligent workflow automation, I quickly adapt to new technologies and business domains. I thrive on turning complex challenges into scalable, user-friendly solutions that create real business impact.

SKILLS

Programming: JavaScript, TypeScript, Python, PHP, Laravel

Frontend: React, Next.js, HTML, CSS, Tailwind CSS

Backend: Node.js, Express, FastAPI, Flask, Laravel

AI & Automation: LangGraph, Agentic AI Design, Workflow Automation, Business Process Automation, n8n, Make.com, REST APIs, Webhooks, JSON

Enterprise Platforms: SharePoint Online, Microsoft Graph API, Power Automate, Power Apps, Microsoft Teams, Microsoft 365, Copilot Studio, Zoho CRM, Zoho Books, Zoho Projects

Databases: PostgreSQL, MongoDB, MySQL, Supabase

DevOps & Tools: Docker, Git, GitHub, Postman

Security (Supporting): Burp Suite, Metasploit, Nmap, Wireshark, SIEM platforms, OWASP Top 10, Malware Analysis

EXPERIENCE

Associate AI Automation Engineer — TechD Cybersecurity Limited

Jan 2026 – Present

Progressed from an initial Technical Delivery & SOC Analyst role into ownership of AI engineering and enterprise workflow automation initiatives.

- **AI Agent Engineering:** Architected and built an agentic AI assistant for Microsoft Teams using LangGraph and FastAPI, giving the Business Development team natural-language access to live SharePoint CRM data (pipeline, quotes, invoices, targets) and on-demand email drafting, with conversation memory maintained across a Teams thread.
- **Architecture & Extensibility:** Designed a provider-agnostic LLM abstraction layer and a tool-registry pattern (search_client, search_quote, search_funnel, search_invoice, draft_email) so new capabilities and LLM providers can be added through configuration rather than rework; implemented a layered API → Agent → Services → Repositories → Graph client architecture with typed Pydantic models, structured logging, and retry/backoff handling.
- **Enterprise Portal Development:** Delivered the PMO Lifecycle Automation Platform — a custom web portal integrated with Microsoft Graph and SharePoint — replacing a manual, email-driven project handover process between Business Development and four delivery departments (SOC, VAPT, GRC, Training) with a structured, metadata-driven intake flow featuring auto-fill, auto-fetch of account owners via Graph directory lookup, and dynamic department-specific fields.
- **Workflow Orchestration:** Engineered two Power Automate flows (Deal-Won Intake, Department Notification) that automatically create tracker records from a won-deal signal and route formatted handover notifications to the correct delivery department without manual emailing.
- **Business Process Automation Platform:** Built the Business Development Lifecycle Automation solution on SharePoint, Microsoft Forms, and Power Automate — six production cloud flows synchronizing quote approvals, deal closure, invoicing, and daily MIS reporting across Sales, Finance/CFO, and Delivery, backed by a cross-list data model joined on a shared Quote ID.
- **Technical Documentation:** Authored enterprise-grade PRDs, FRDs, and implementation/reference guides defining scope, architecture, functional requirements, and success metrics for each automation initiative — written to be directly implementation-ready for engineering, security, and administrator review.
- **SOC Foundations (Early Tenure):** Performed real-time alert triage and incident classification in a live SOC environment; automated recurring compliance-reporting workflows, reducing manual reporting overhead by an estimated 40%.

Cybersecurity Intern — Tinkering Hub, Parul University

Dec 2024 – Apr 2025

- Engineered a real-time malware detection platform (Python) with behavioral and signature-based scanning and automated quarantine, deployed on live university infrastructure with 90%+ threat identification accuracy.
- Built a phishing-detection browser extension performing DOM analysis, URL entropy checks, and real-time reputation lookups (VirusTotal, AbuseIPDB), fully client-side with zero added latency.

Cybersecurity Intern — SkillVertex

Dec 2024 – Jan 2025

- Executed end-to-end VAPT engagements across web and network targets (Nmap, Burp Suite, Metasploit), identifying critical vulnerabilities including SQLi, XSS, and IDOR, and produced CVSS-scored, developer-ready remediation reports.

ENTERPRISE AUTOMATION & AI PROJECTS

BD AI Assistant — Agentic AI Teammate for Microsoft Teams

LangGraph • FastAPI • Microsoft Graph API • Microsoft Teams • Pydantic

- Problem: Business Development data lived across five separate SharePoint lists, requiring manual opening, filtering, and cross-referencing to answer routine pipeline questions.
- Solution: A single LangGraph agent, invoked from a FastAPI backend on new Teams messages, reasons over intent, selects from a read-only tool set to query SharePoint via Graph, and replies inline in the same Teams thread — with conversation state carried across follow-up turns.
- Engineering decisions: Kept V1 deliberately scoped (read-only data access, draft-only email) while designing the tool registry, LLM provider interface, and checkpoint-ready state so V2 capabilities (CRUD, calendar, send) are additive, not a rewrite; defined explicit success metrics (grounded-answer rate, response latency, context-carry success) up front.

PMO Lifecycle Automation Platform — Custom Portal + Graph + Power Automate

Microsoft Graph API • SharePoint Online • Power Automate • Custom Web Application

- Problem: Handover from a closed deal to project delivery relied on manual emails and spreadsheet updates across four delivery departments, with no single source of truth for initiation status.
- Solution: Built a guided, multi-step Project Initiation Portal that reads verified client data from a SharePoint Client Master via Microsoft Graph, auto-fills contact and account-owner details, adapts its fields to the department selected, and writes the finalized record into a central Main Tracker.
- Impact: Eliminated manual re-keying of client data and email-based handovers; gave PMO and management a real-time, centralized view of initiation status across all departments.

Business Development Lifecycle Automation — SharePoint + Forms + Power Automate

SharePoint Online • Microsoft Forms • Power Automate • Office 365 Outlook • Excel Online

- Problem: No connected record linked a quotation, its pricing approval, the resulting deal, and its invoices, leaving approved quotes untracked and payments unreconciled against opportunities.
- Solution: Designed a three-list data model (Quote Approvals → Funnel → Invoices) joined on a generated Quote ID, with six Power Automate flows enforcing business rules — approval routing, deal-won notification to Finance and Delivery, invoice-to-opportunity reconciliation, and an automated daily MIS summary email.
- Impact: Replaced disconnected manual tracking with a synchronized, auditable pipeline from quotation through payment collection, with automated leadership reporting.

ADDITIONAL PROJECTS

- **RansomeWatch & SecureScope** — Laravel-based malware detection/incident-management platform with real-time scanning and an extensible rule engine; Python/Flask OWASP Top 10 scanner with a modular plugin architecture and React frontend.
- **Endpoint & Phishing Defense** — Windows endpoint tool using heuristic, behavior-based threat classification (<2% idle CPU); fully client-side browser extension combining URL entropy analysis and brand-spoofing detection.

EDUCATION

B.Tech, Computer Science Engineering — Cybersecurity Specialization

2022 – 2026

Parul University, Vadodara, Gujarat

CERTIFICATIONS & TRAINING

AI Automation Explorer – Make Academy (Jul 2026)

C3SA – Certified Cloud Security Analyst (CyberWarfare Labs)

Google Cloud Cybersecurity Professional Certificate

AWS Cloud Essentials

Ethical Hacking Essentials (EC-Council)

Kali Linux (Offensive Security)